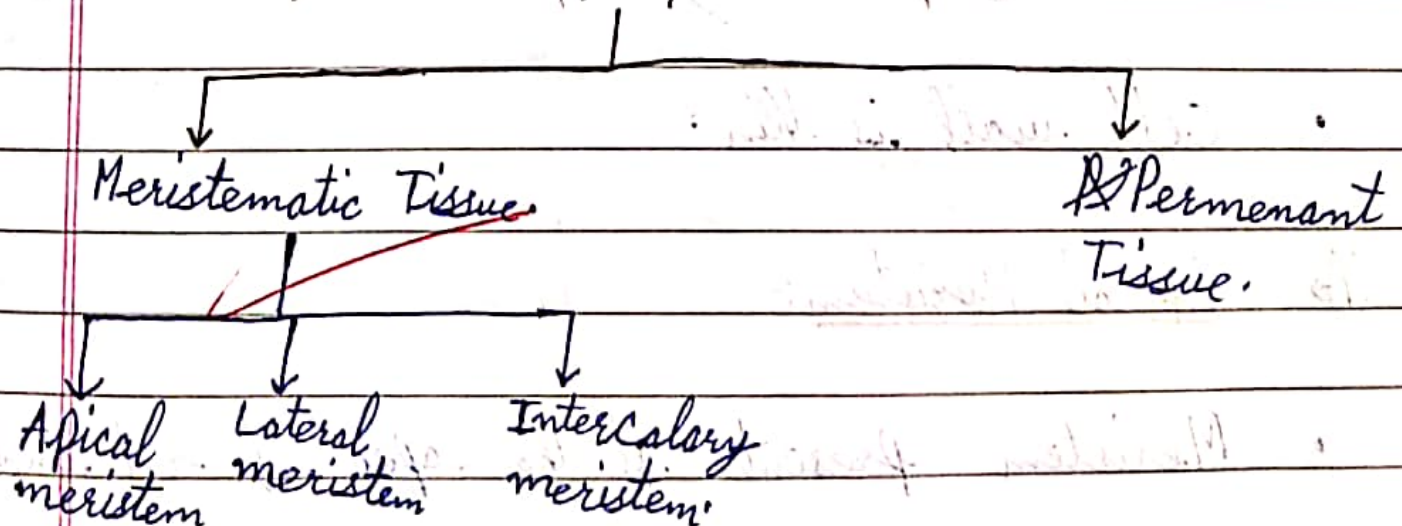


Tissue

- Tissue: Group or collection of cells.
 - They perform a specific function.
 - So that division of labour occurs.
 - Formation of an organisation.
i.e. cell → tissue → Organ → Organ System → Organism.
 - Cells in a tissue can be similar or dissimilar
- ⇒ Plant tissue, Animal tissue:

The tissues of plants are known as Plant Tissue.

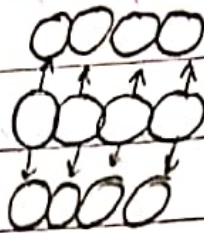
→ Classification of plant tissues



⇒ Meristematic Tissue:-

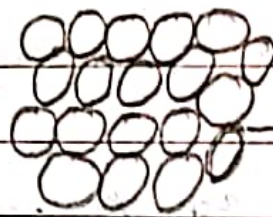
- A tissue which is under a continuous state

of division.



- The cells of meristematic tissue are oval, spherical, isodiametric.

- Cells have intercellular spaces.



Intercellular Space.

- Dense cytoplasm and prominent nucleus.

- Cell wall is thin.

(1) Apical Meristem:-

- Meristem present at the apex of root and

shoot.

- Root $\rightarrow$ RAM (Root apical meristem)
- Shoot $\rightarrow$ SAM (Shoot apical meristem).

$\rightarrow$ It increases length of plants.

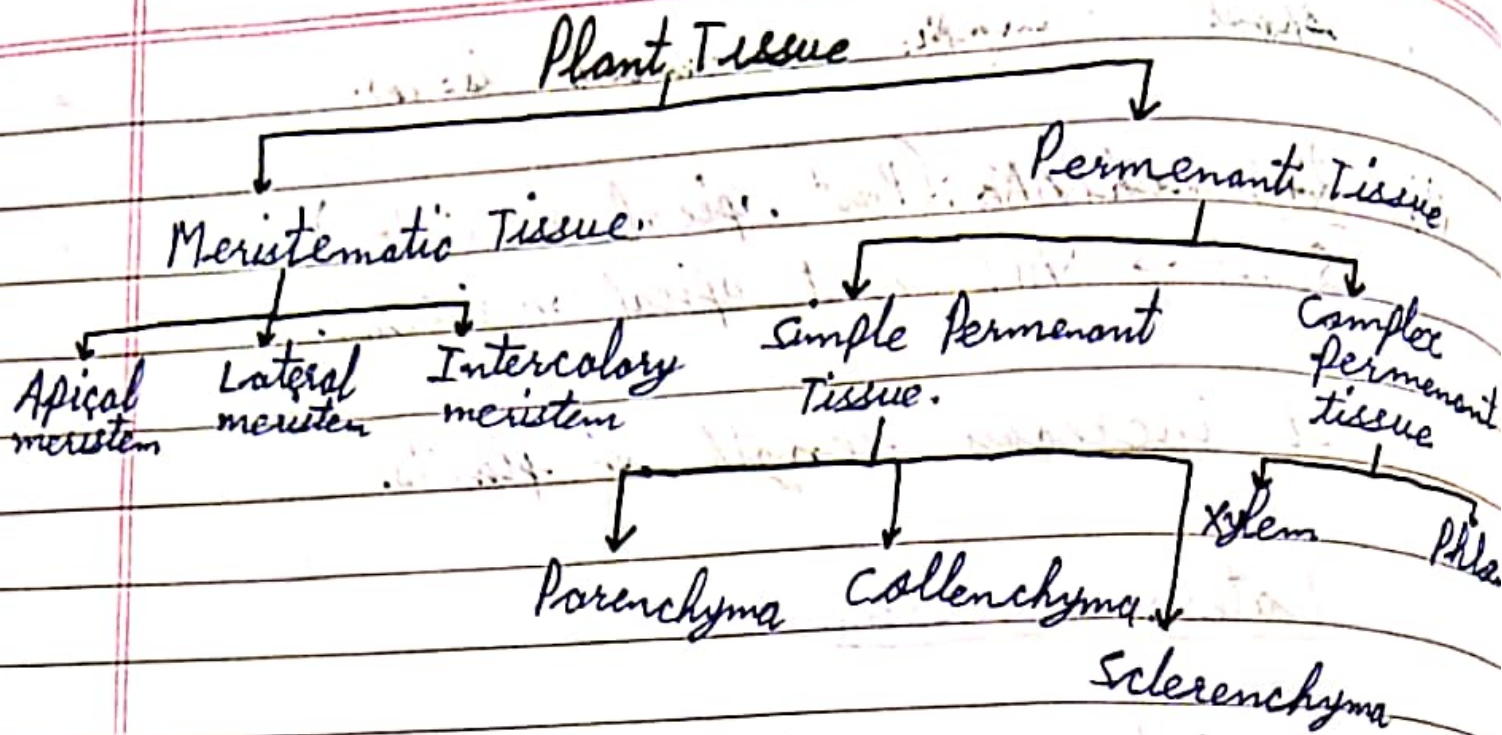
(2) Lateral meristem.

- Meristem present on lateral sides.
- Increases girth of plant.

(3) Intercalary Meristem.

- Meristem present at node and at base of leaf.

- Increases length of organ.



⇒ Permanent Tissue:

- It is a group of fully grown mature cells, which have lost the capacity for division.

⇒ Types.

Simple Permanent tissue.

- Tissue made up of only one type of cell.

Complex/compound permanent tissue.

- Tissue made up of more than one type of cell.

⇒ Types of Simple Permanent Tissue:-

① Parenchyma:-

- Living cells.
 - Thin cell wall.
 - With intercellular spaces.
- They perform function of temporary storage.
- If parenchyma having chlorophyll (chloroplast) it is then called as "chlorenchyma".
- In aquatic plants parenchyma is modified to store air → known as "aerenchyma".

② Collenchyma:-



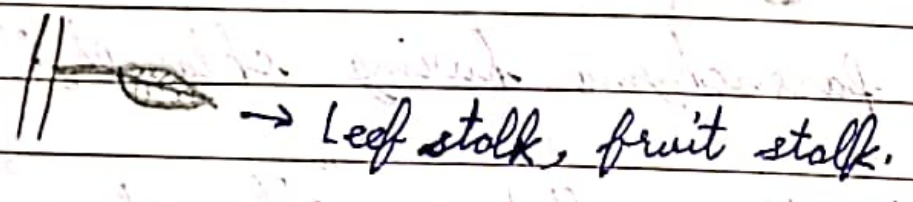
⇒ Exchange of substance

- Spherical or polygonal = shape
- Intercellular spaces are lost, due to deposition of pectin

→ ~~Living tissue~~

→ Function:-

- Provides flexibility to various parts of plants.



- Thus it allows bending of plant parts like tendrils, [3] and climber.

③ Sclerenchyma:-

⇒ It is the tissue which makes plant hard.

⇒ Therefore it provides mechanical support to

plant.

- ⇒ • The cells of this tissue are dead → due to deposits of lignin.
- ⇒ • The cells of this tissue are very thick walled, such that there is no intercellular space.



* The tissue is present in:

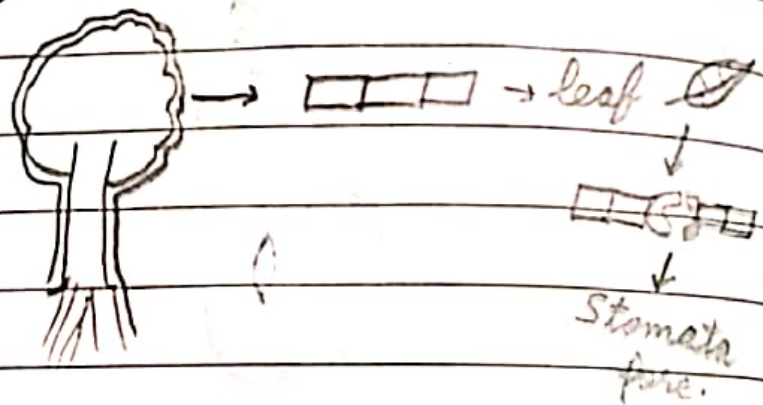
- Hard covering of seeds and nut.
- Coconut husk.
- Vascular bundle - Xylem and phloem.

- Mechanical strength:

- A structural frame work that allows to with stand ~~ph~~ physical forces like, wind, gravity.

⇒ NCERT - Booster:

- Outer most layer of plant is known as epidermis.



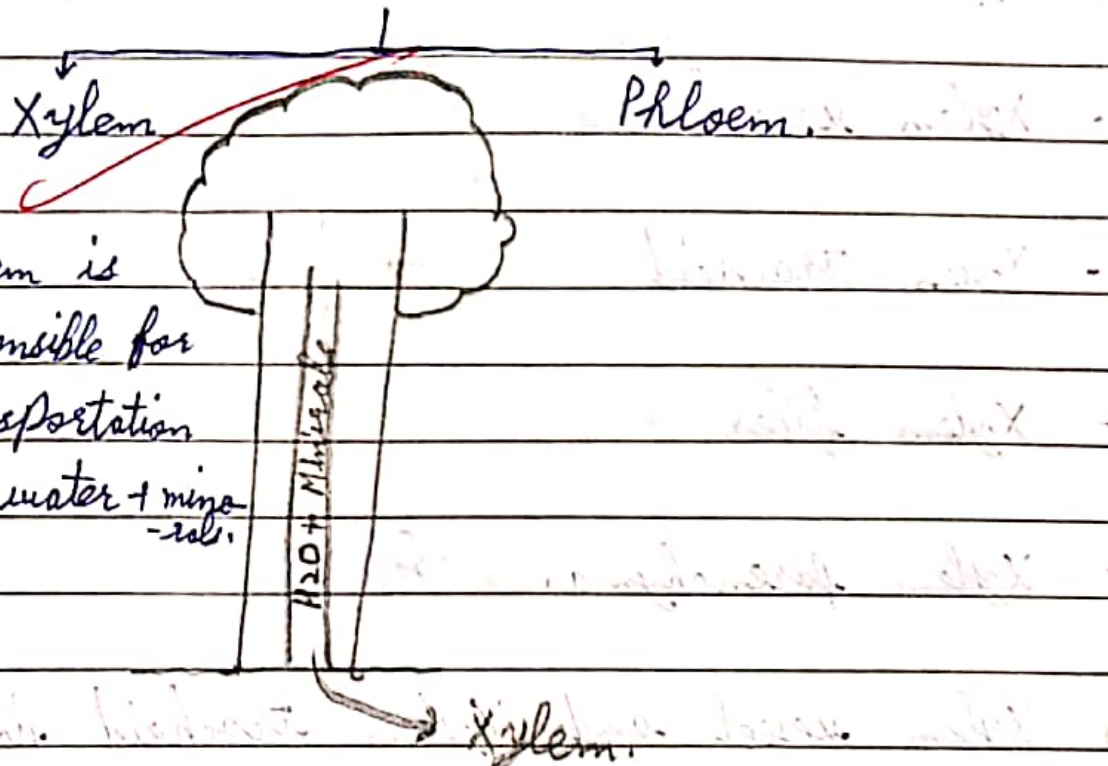
- Epidermal cells are compactly arranged. helps in exchange of gases and water

- In stem and older branches, epidermis gets modified and form a protective covering known as "cork". → (Cork is dead) → It has deposition of

suberin.

⇒ Compound/Complex permanent tissue:-

- This type of tissue is made up of more than one type of cells.



- Xylem is responsible for transportation of water + minerals.

- Components of Xylem:-

- Xylem vessels = Dead. } Main elements for transportation
- Xylem tracheid = Dead.
- Xylem Fiber = Dead
- Xylem Parenchyma = ~~Dead~~ Living.

→ Xylem and Phloem are known conducting elements, thus they constitute "vascular bundle" of a plant.

→ Xylem consists of:

- Xylem vessel →



- Xylem tracheid →



- Xylem fiber →



- Xylem parenchyma →



⇒ Xylem vessel and Xylem tracheid have thick walls and are dead at maturity.

→ Xylem parenchyma stores food.

→ Xylem fiber provides mechanical support.

⇒ Phloem

It is the tissue which transports food from leaves to all other parts of plant.

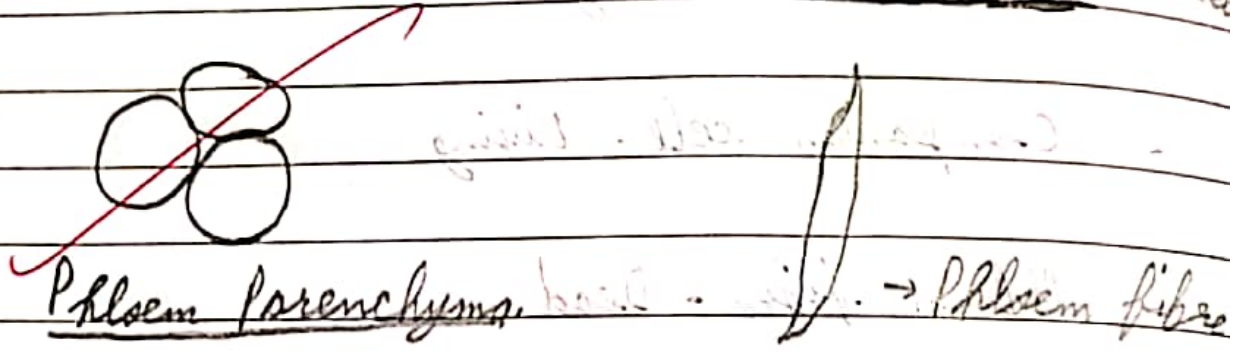
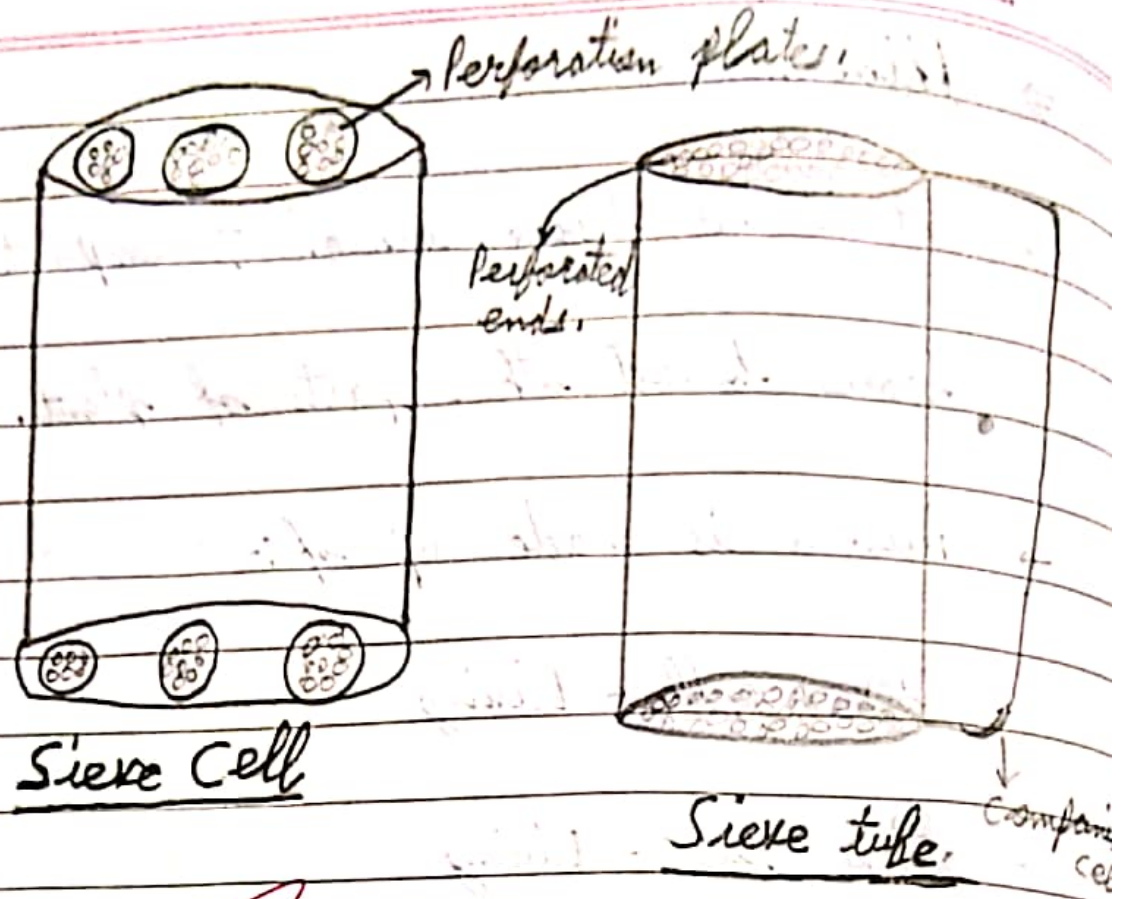
→ Phloem is made up of:-

- Sieve cell - Living
- Sieve tube - Living
- Companion cell - Living
- Phloem fibre - Dead
- Phloem Parenchyma - Living

⇒ Sieve tubes are the tubular cells, with perforated walls.

- ~~Phloem fibre~~ is dead, rest of the cells are living.

Seen
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1) → Sieve cell have perforation restricted to a particular area called perforation plate.

→ Sieve cell is shorter, compared to sieve tube.

2) Sieve Tube ~ These tubular, elongated, placed

end-to-end

→ End walls have tiny pore.

3) Companion Cell:-

→ These are closely associated with Sieve tube.

4) Phloem Parenchyma.

- These cells store food and help in conduction of nutrients.

5) Phloem Fibre

- ~~These~~ They are dead and provide support.

*→ Phloem helps in transportation of food material from leaves to all other parts including storage organs.

*→ Transportation of food is bi-directional, i.e; both upward and downward.

⇒ Animal Tissue:-

- The tissues present in the body of animals.

⇒ Types of tissues:-

- 1) Epithelial tissue
- 2) Connective tissue
- 3) Muscular tissue
- 4) Nervous tissue

1) ⇒ Epithelial Tissue:-

It forms outermost covering and inner lining of body.

- Cells of this tissue are tightly packed, and form a continuous sheet.

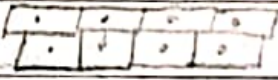


→ Normally they have centrally placed Nucleus.

- There are no intercellular spaces in between.

⇒ Types:

1) Squamous epithelium = 

2) Stratified epithelium = 

3) Cuboidal epithelium = 

4) Columnar epithelium = 

5) Ciliated epithelium = 

6) Glandular epithelium.

1) Squamous Epithelium:

- These cells are extremely thin, and they are flat.

- Found in the regions of selective transport.
- e.g. Blood Vessel, Lung Alveoli.

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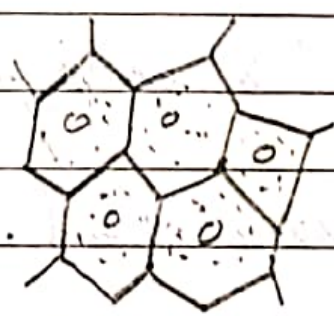
★ ⇒ Epithelial Tissue:

- This tissue grows on some other tissue.
- The cells of this tissue are, flat, cuboidal, columnar or polygonal.
- There is no direct blood supply to tissue.
- The cells ^{of this} ~~are~~ tissue does not have inter cellular space.

⇒ Epithelial tissue is of following types:-

- 1) - Squamous epithelium
- 2) - Cuboidal epithelium
- 3) - Columnar epithelium
- 4) - Ciliated epithelium
- 5) - Glandular epithelium

1) Squamous Epithelium:-



- These are thin, flat cells.
- They fit into each other like tiles.
- Present in the regions of selective transport.
- Many layers of squamous epithelium form stratified

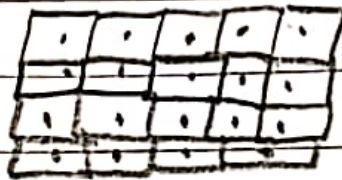
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squamous epithelium.

e.g. Skin.

2) Cuboidal Epithelium:-

- This tissue has cube like cell.



- Iso diametric.
 - Closely fitted into each other.
- • Present in Salivary gland, Thyroid gland, Gland.

- These cells are protective in function.

3) Columnar epithelium:-

- These cells are tall or pillar like.

*⇒ They help in absorption of nutrients.

- Location :- Intestines
 - Gall bladder
 - Stomach.

4) Ciliated Epithelium :-

- Cube like or columnar.
- They have small hair-like structure known as cilia.



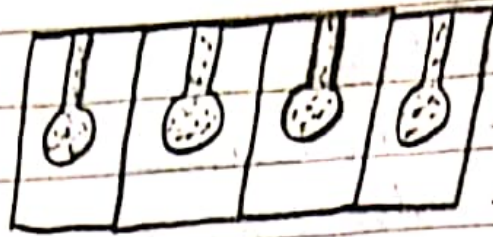
- Location :- Nasal Passage
 - Wind pipe.

- Helps in the movement of Mucus.

5) Glandular Epithelium :-

- Cells of glandular epithelium help in secretion.

of substance.



- Location = Skin (Sweat Gland).

⇒ Connective Tissue

- The tissue which connects and binds different tissues and organs.

→ Types:

1) Adipose Tissue

2) Tendons.

3) Ligament

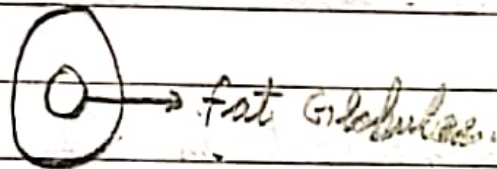
4) Cartilage

5) Bone.

6) Blood.

1) Adipose Tissue.

- The cells of this tissue are large, oval, contains fat globules.



- Present below skin
- Present around heart, liver and eye ball.

2) Tendon

- Bundle of white fibers.
- It joins muscle with bones.

3) Ligament

- Bundle of yellow fibers.

→ It joins bone with bone.

4) Cartilage

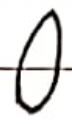
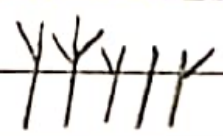
- It forms supporting frame work.
- It is softer than bones.
- It is elastic because of protein "Chondrin".
- It is present in nose, ear, wind pipe.

5) Bone

- Hardest connective tissue.
- It has a cavity, the cavity is filled with bone marrow.
- It provides surface for attachment of muscle.
- It forms supporting frame work of body.

⇒ Muscular Tissue

- The tissue which helps in movement of body parts.
- There are three types of muscles.

Striated Muscle. OR Skeletal Muscle.	Unstriated Muscle OR Smooth Muscle.	Cardiac Muscle OR Muscles of Heart
• Voluntary i.e; controlled by our will. • Long and cylindrical and unbranched.	• Involuntary. i.e; not controlled by our will. • Spindle shaped, Unbranched.	• Involuntary i.e; not controlled by our will. • Cylindrical and branched.
		
• Multinucleate • Attached to bones	• Uninucleate • Found in walls of internal organ.	• Uninucleate. • Found exclusively walls of heart.

⇒ Nervous Tissue

- It is tissue which is specialized for receiving and transmitting message in our body.
- It has specialized cells called "neuron".
- This tissue controls all the activities of body.
- A neuron or nerve cell has three parts:-

1) Cell body or cyton

2) Dendron

3) Axon.

